

■ 2.4.5 Applying Distributive Laws

There are a number of properties that you often need to define when you treat an object like f as an “operator”. Section 2.2.18 discussed how to define f to be associative and commutative. This section discusses making f *distributive*.

If f is distributive over **Plus**, then an expression like $f[a + b]$ can be “expanded” to give $f[a] + f[b]$. The function **Expand** performs this kind of expansion using the distributivity of operators like **Times**.

You can use the function **Distribute** to expand expressions involving any operator f , and to implement the distributive law for this operator over operators like **Plus**.

This “distributes” f over $a + b$.

```
In[1]:= Distribute[ f[a + b] ]
Out[1]= f[a] + f[b]
```

Here is a more complicated example.

```
In[2]:= Distribute[ f[a + b, c + d] ]
Out[2]= f[a, c] + f[a, d] + f[b, c] + f[b, d]
```

The result is the analogue of the expansion of this product.

```
In[3]:= Expand[ (a + b) (c + d) ]
Out[3]= a c + b c + a d + b d
```

You can tell **Distribute** to distribute over operators other than **Plus**. The result contains all possible pairs of arguments.

```
In[4]:= Distribute[ f[{a, b}, {c, d}], List ]
Out[4]= {f[a, c], f[a, d], f[b, c], f[b, d]}
```

In this case you can get a similar result using **Outer**.

```
In[5]:= Flatten[ Outer[f, {a, b}, {c, d}] ]
Out[5]= {f[a, c], f[a, d], f[b, c], f[b, d]}
```

When you apply the distributive law to two operators f and g , the results you get do not necessarily have to contain the same operators. You can tell **Distribute** exactly which operators you want to distribute over, and exactly what operators to insert in the results.

```
In[6]:= Distribute[ f[g[a, b], g[c, d]], g, f, fp, gp ]
Out[6]= fp[gp[a, c], gp[a, d], gp[b, c], gp[b, d]]
```

This is the kind of distribution that you need to do for an operator $\exp$ that is analogous to exponentiation.

```
In[7]:= Distribute[ exp[a + b + c],
                  Plus, exp, Times, exp ]
Out[7]= exp[a] exp[b] exp[c]
```

<code>Distribute[f[a + b + ..., ...]]</code>	distribute f over sums to give $f[a, ...] + f[b, ...] + \dots$
<code>Distribute[f[args], g]</code>	distribute f over any arguments which have head g
<code>Distribute[expr, g, f]</code>	distribute only when the head is f
<code>Distribute[expr, g, f, gp, fp]</code>	distribute f over g to give fp and gp , respectively

Applying distributive laws.